

Project Planning Report: Optimisation Methods for Variational Inference

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Background and Problem Intuition

Bayesian inference provides principled uncertainty quantification but requires computing intractable posterior distributions. For most real-world models, the posterior integral,

$$p(\boldsymbol{\theta} \mid \mathbf{y}) = \frac{p(\mathbf{y} \mid \boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\mathbf{y})},$$

cannot be evaluated analytically. Variational Inference (VI) converts this inference problem into an optimisation problem, making it directly amenable to the gradient-based methods in this course.

Instead of computing the true posterior, VI finds the “best” approximation $q(\boldsymbol{\theta})$ from a tractable family by maximising the Evidence Lower Bound (ELBO). This transforms intractable integration into tractable optimisation—I think this is a natural fit for an optimisation course project.

Key Notation

Symbol	Meaning
$\boldsymbol{\beta}$	Regression coefficient vector
$\mathbf{u}$	Random effects vector (hierarchical model)
$\boldsymbol{\varepsilon}$	Observation noise vector
τ_e	Observation-noise precision
τ_u	Random-effects precision
$\boldsymbol{\mu}_\beta$	Mean of variational posterior $q(\boldsymbol{\beta})$
$\boldsymbol{\Sigma}_\beta$	Covariance of variational posterior $q(\boldsymbol{\beta})$

Optimisation Problem Formulation

Decision Variables

For Bayesian linear regression with p predictors, the variational parameters (decision variables) are:

- $\boldsymbol{\mu}_\beta \in \mathbb{R}^p$: posterior mean of regression coefficients
- $\boldsymbol{\Sigma}_\beta \in \mathbb{R}^{p \times p}$: posterior covariance (symmetric positive definite)
- $a_e, b_e > 0$: shape and rate parameters of the variational Gamma factor for precision

Here, a_e and b_e parameterise $q(\tau_e) = \text{Gamma}(a_e, b_e)$; the model prior on τ_e is specified in the Test Case section.

Total dimension: $d = p + \frac{p(p+1)}{2} + 2 = 11$ variables for $p = 3$ predictors.

Objective Function

Maximise the Evidence Lower Bound (ELBO):

$$\text{ELBO}(\boldsymbol{\phi}) = \mathbb{E}_q[\log p(\mathbf{y}, \boldsymbol{\beta}, \tau_e)] + H(q) \quad (1)$$

$$= \mathbb{E}_q[\log p(\mathbf{y}, \boldsymbol{\beta}, \tau_e)] - \mathbb{E}_q[\log q(\boldsymbol{\beta}, \tau_e)]. \quad (2)$$

where $\boldsymbol{\phi} = (\boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta, a_e, b_e)$ are the variational parameters.

Constraints

- $\boldsymbol{\Sigma}_\beta \succ 0$: Covariance matrix must be positive definite
- $a_e, b_e > 0$: Gamma parameters must be positive
- Optional: Minimum variance constraints $\sigma_{\beta_j}^2 \geq \sigma_{\min}^2$ to prevent underdispersion

These constraints can be handled via parameterisation (e.g., Cholesky decomposition for $\boldsymbol{\Sigma}_\beta$) or penalty methods.

Computational Characteristics

- Gradient computation: $O(np^2)$ per evaluation (dominated by matrix operations)
- Hessian computation: $O(np^2 + d^3)$ for Newton's method
- Parameters exhibit coupling (changing $\boldsymbol{\mu}_\beta$ affects optimal $\boldsymbol{\Sigma}_\beta$)
- Must handle special functions: digamma $\psi(\cdot)$, trigamma $\psi'(\cdot)$ for Gamma distribution

Test Case: Bayesian Linear Regression

The test case uses Bayesian linear regression with:

$$\mathbf{y} \sim \mathcal{N}(\mathbf{X}\boldsymbol{\beta}, \tau_e^{-1}\mathbf{I}_n) \quad (\text{likelihood}) \quad (3)$$

$$\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, \lambda_\beta^{-1}\mathbf{I}_p) \quad (\text{prior on coefficients}) \quad (4)$$

$$\tau_e \sim \text{Gamma}(\alpha_e, \gamma_e) \quad (\text{prior on precision}) \quad (5)$$

Mean-field approximation:

$$q(\boldsymbol{\beta}, \tau_e) = q(\boldsymbol{\beta}) q(\tau_e) \quad (6)$$

$$q(\boldsymbol{\beta}) = \mathcal{N}(\boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta) \quad (7)$$

$$q(\tau_e) = \text{Gamma}(a_e, b_e) \quad (8)$$

Variational parameters: $\boldsymbol{\phi} = (\boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta, a_e, b_e)$ with dimension $d = p + \frac{p(p+1)}{2} + 2$

For $p = 3$ predictors: $d = 3 + 6 + 2 = 11$ parameters to optimise.

Overview of Optimisation Methods

Methods considered

- CAVI: Domain-specific baseline exploiting problem structure
- Gradient Ascent: First-order baseline method using gradient information and line search
- Newton's Method: Second-order method with quadratic local convergence near a well-conditioned optimum
- BFGS: Quasi-Newton method that approximates Hessian information; practical general-purpose method for smooth unconstrained problems

Baseline Method: CAVI¹

Coordinate Ascent Variational Inference (CAVI) is VI-specific. It updates each variational parameter block sequentially using closed-form solutions:

$$\boldsymbol{\Sigma}_\beta \leftarrow \left(\frac{a_e}{b_e} \mathbf{X}^T \mathbf{X} \right)^{-1} \quad (9)$$

$$\boldsymbol{\mu}_\beta \leftarrow \frac{a_e}{b_e} \boldsymbol{\Sigma}_\beta \mathbf{X}^T \mathbf{y} \quad (10)$$

$$a_e \leftarrow \alpha_e + \frac{n}{2} \quad (11)$$

$$b_e \leftarrow \gamma_e + \frac{1}{2} [\|\mathbf{y} - \mathbf{X}\boldsymbol{\mu}_\beta\|^2 + \text{tr}(\mathbf{X}^T \mathbf{X} \boldsymbol{\Sigma}_\beta)] \quad (12)$$

Properties: Monotonic ELBO increase ², linear convergence, $O(np^2)$ per iteration. Provides analytical benchmark for validation.

¹Blei, Kucukelbir, and McAuliffe (2017), Variational Inference: A Review for Statisticians (Journal of the American Statistical Association).

²Monotonic ELBO increase means the objective is non-decreasing across iterations: $\mathcal{L}^{(t+1)} \geq \mathcal{L}^{(t)}$. In CAVI, each coordinate update maximises the ELBO with respect to one variational factor while holding the others fixed, so each step cannot decrease the ELBO.

Mapping ENEL445 Methods to VI Problem

1. Parameter Ascent (Course: gradient descent)

- Standard iterative update: $\phi^{(t+1)} = \phi^{(t)} + \alpha_t \nabla \text{ELBO}(\phi^{(t)})$
- Line search from Chapter 3.2 (bisection/Wolfe conditions)
- Stopping criteria: $\|\nabla \text{ELBO}\| < \epsilon$
- Convergence characteristics: assuming it follows a linear rate

2. Newton's Method

- Standard Newton update: $\phi^{(t+1)} = \phi^{(t)} + \alpha_t [\nabla^2 \text{ELBO}]^{-1} \nabla \text{ELBO}$
- Requires computing and inverting the Hessian matrix
- Computational cost: $O(d^3)$ per iteration for Hessian inversion
- Convergence characteristics: unknown.
- Challenge: Hessian may not be positive definite away from the optimum

3. BFGS Quasi-Newton (one of five required methods from Project 2)

- BFGS update formula builds inverse Hessian approximation
- Uses gradient differences: $\mathbf{s}_k = \phi^{(k)} - \phi^{(k-1)}$, $\mathbf{y}_k = \nabla \text{ELBO}(\phi^{(k)}) - \nabla \text{ELBO}(\phi^{(k-1)})$
- Computational cost: $O(d^2)$ per iteration
- Convergence characteristics: unknown.

Comparison will reveal whether general optimisation methods can compete with or outperform problem-specific algorithms

Solution Plan

Phase 1: Data Generation and Baseline

Data simulation I (complete):

- Generate $n = 100$ observations, $p = 3$ predictors
- Design matrix $\mathbf{X} \sim \mathcal{N}(0, 1)$, true coefficients $\beta_{\text{true}} = [1, -0.5, 2]^T$
- Noise: $\epsilon \sim \mathcal{N}(0, 0.5^2)$

Mathematical foundations (in progress):

- Complete ELBO derivation for Bayesian linear regression
- CAVI closed-form update equations
- Gradient expressions: $\nabla_{\mu_\beta} \text{ELBO}$, $\nabla_{\Sigma_\beta} \text{ELBO}$, $\nabla_{a_e} \text{ELBO}$, $\nabla_{b_e} \text{ELBO}$
- Hessian block structure: diagonal and mixed blocks
- BFGS update formulation with Woodbury identity

Derivations to be documented in `vi_optimisation_solution.pdf` (or an appendix to the report?)

Phase 2: Implementation (in progress)

1. Data Generation (5%)

- Simulate data: $n = 100$ observations, $p = 3$ predictors
- Generate $\mathbf{X}$ from standard normal, $\boldsymbol{\beta}_{\text{true}} = [1, -0.5, 2]^T$
- Add noise: $\epsilon \sim \mathcal{N}(0, 0.5^2)$

2. Baseline Implementation: CAVI (15%)

- Implement closed-form coordinate updates
- Track ELBO convergence
- Stopping criterion: $|\text{ELBO}^{(t+1)} - \text{ELBO}^{(t)}| < 10^{-6}$

3. Method 1: Gradient Ascent (15%)

- Implement ELBO gradient computation
- Line search: bisection method or Wolfe conditions
- Stopping criterion: $\|\nabla \text{ELBO}\| < 10^{-6}$
- Track iteration count and ELBO values

4. Method 2: Newton's Method (0%)

- Implement Hessian computation (block structure)
- Regularisation: can I add $\lambda \mathbf{I}$ if Hessian not positive definite?
- Line search for step size α_t
- Track convergence and computational cost

5. Method 3: BFGS Quasi-Newton (0%)

- Initialise inverse Hessian approximation: $\mathbf{B}_0 = \mathbf{I}_d$
- Track superlinear convergence

6. Comparison and Analysis (5%)

- Compare convergence speed (iterations to 10^{-6} tolerance)
- Compare computational cost (time per iteration, total time)
- Compare solution quality (posterior approximation accuracy)
- Analyse when each method performs best

Success Metrics

1. Convergence: All methods reach ELBO optimum within tolerance
2. Correctness:
 - CAVI correctness: Gradient/Newton/BFGS implementation is correct if it reaches the same ELBO optimum / variational fixed point as CAVI (same VI objective, same variational family)
 - MCMC correctness: VI is assessed against MCMC as a posterior-accuracy benchmark (means, intervals, calibration), not expected to match exactly because VI is approximate
3. Performance hierarchy:
 - CAVI likely fastest for this conjugate problem
 - MCMC/Gibbs expected slowest computationally, but best for posterior uncertainty accuracy (gold-standard benchmark, not a direct ELBO-optimiser competitor)

Progress To Date

Work in progress using an iterative / Agile method

1. Problem formulation: Draft optimisation problem specified with decision variables, objective function (ELBO), and constraints
2. Mathematical derivations³:
 - Draft ELBO derivation from first principles
 - Draft CAVI algorithm with closed-form updates
 - Analytical gradient expressions for all 11 variational parameters (unverified)
 - Potential Hessian block computations for Newton's method
 - Attempt BFGS quasi-Newton formulation
 - Attempt at Penalty method formulation for constrained optimisation
3. Validation strategy: Established CAVI as analytical benchmark for correctness checking

Preliminary Results

Implementation is pending, the mathematical analysis hints⁴

³Desperately in need of some kind of peer review

⁴I havent been over this enough to make any claim of completeness

Next Steps

1. Implement CAVI baseline in Python (Week 7)
2. Implement parameter ascent with line search (Week 7)
3. Implement Newton's method with regularisation (Week 8)
4. Implement BFGS quasi-Newton (Week 8)
5. Monte Carlo comparison: convergence iterations, wall-clock time, ELBO values (Week 8-9)
6. Final report with performance analysis (Week 12)

Alignment with Course Requirements

The VI problem is meant to represent real computational challenges, making the comparison of optimisation methods valuable, relevant, and interesting.

Timeline

TBD

Summary

I've set up the optimisation problem and derived the math. The next step is to code four algorithms—CAVI, gradient ascent, Newton, BFGS—and see which one wins, and why.